NLP Assignment 1: Word2Vec

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1 Experimental Settings

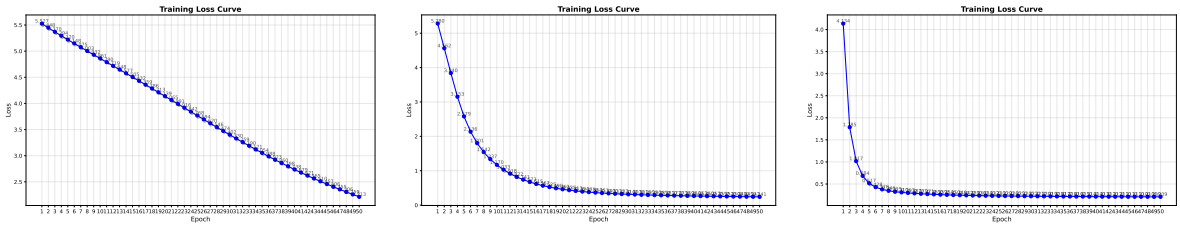
为了客观的评估不同超参数训练出的 Word2Vec 模型能否正确的捕获自然语言中的相关性信息并嵌入到高维空间中，我们设计了如下指标来客观地评估：

- **Training Curves**: 训练过程中的 Loss 下降曲线，验证模型训练的稳定性。
- **Similar Results**: 测试给定的不同类型 token（常见结构助词、简单单词、复杂单词）在词表中最相关的其他 token，检查模型是否捕获到了正确的语义相关性。
- **可视化**: 从词表中随机采样 200 个 token，对其高维向量进行降维 (PCA, tSNE) 并可视化
- **Word Analogy**: 词类比任务, word1 − word2 + word3 =?

2 `learning_rate`

- `learning_rate`: 0.001, 0.005, 0.0001
- `embed_size`: 200
- `epoch`: 50

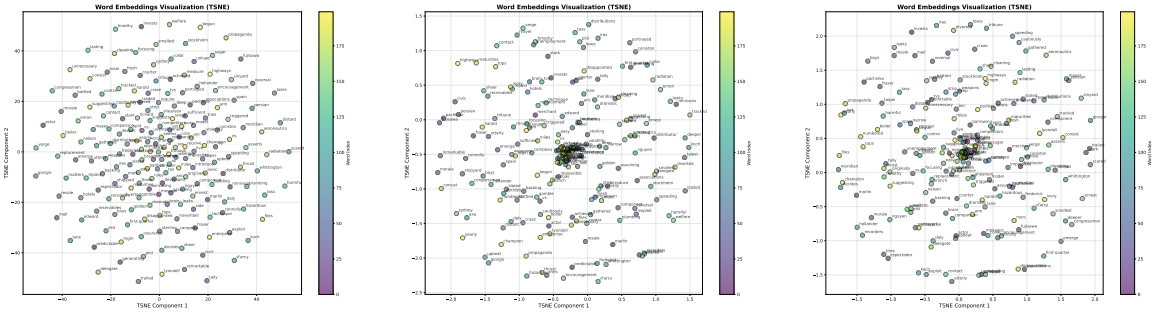
2.1 Training Curves



(a) LR = 0.0001 (b) LR = 0.001 (c) LR = 0.005

Figure 1: 不同学习率下的 Word2Vec 训练曲线对比

2.2 Visualizations



(a) LR = 0.0001 (b) LR = 0.001 (c) LR = 0.005

Figure 2: 不同学习率下的词向量 t-SNE 可视化结果对比

2.3 Similar Results

2.3.1 Learning Rate: 0.0001

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	franco	rush	spouse	representatives	photos
he	that	the	his	a	to
bank	corp.	million	officer	of	a
responsibility	congressional	task	legal	combined	suggests
eagerness	opportunities	incorporated	boring	candlestick	promotions

Table 1: Top 5 similar words for learning rate 0.0001

2.3.2 Learning Rate: 0.001

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	listening	recall	floor	peddling	he
he	it	had	no	have	that
bank	banking	securities	branch	loans	rating
responsibility	day-to-day	connections	responsible	wolf	shipyard
eagerness	simpler	poorest	unless	jokes	introduce

Table 2: Top 5 similar words for learning rate 0.001

2.3.3 Learning Rate: 0.005

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	assault	venerable	broadly	christie	hazard
he	schools	considered	balloon	oversee	multinational
bank	withdraw	attorney	berlin	tuesday	savaiko
responsibility	defaulted	negotiated	fines	arkansas	inventories
eagerness	integrated	attacked	ernest	township	hosts

Table 3: Top 5 similar words for learning rate 0.005

2.4 Word Analogy

2.4.1 Learning Rate: 0.0001

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	burger	grounds	j.p.	crucial	stand
paris - france + italy	finish	sabotage	bank	day	prices
tokyo - japan + germany	communist	attended	shevardnadze	mercedes	deck
doctor - hospital + teacher	walks	fla	select	relied	companion

Table 4: Top 5 word analogy predictions for learning rate 0.0001

2.4.2 Learning Rate: 0.001

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	pittsburgh	u.s.	rainbow	scientific	include
paris - france + italy	nov.	college	presidency	governments	killed
tokyo - japan + germany	ringing	stock	germans	shearson	midsized
doctor - hospital + teacher	conducted	iran-contra	manuel	mrs.	press

Table 5: Top 5 word analogy predictions for learning rate 0.001

2.4.3 Learning Rate: 0.005

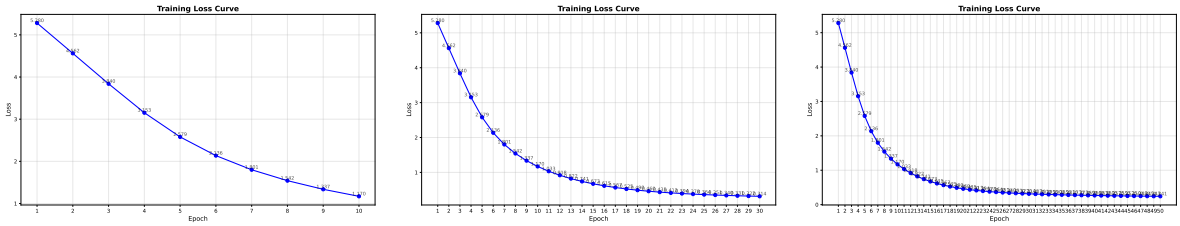
Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	chips	jenrette	fiscal	cautioned	payouts
paris - france + italy	smaller	businessland	everyone	compelling	brouwer
tokyo - japan + germany	reopened	circles	direction	steadily	steven
doctor - hospital + teacher	lama	ounce	silly	1/2-year	possibility

Table 6: Top 5 word analogy predictions for learning rate 0.005

3 num_epochs

- `learning_rate`: 0.001
- `embed_size`: 200
- `epoch`: 10, 20, 30, 40, 50

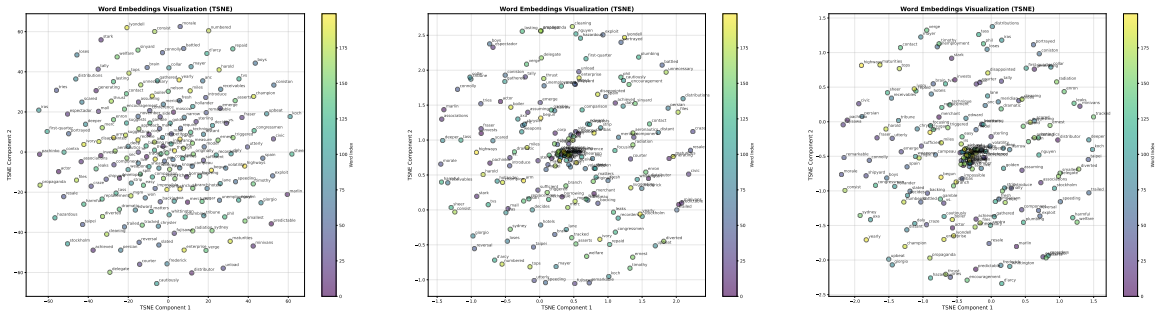
3.1 Training Curves



(a) Epochs = 10 (b) Epochs = 30 (c) Epochs = 50

Figure 3: 不同迭代次数下的 Word2Vec 训练过程对比

3.2 Visualizations



(a) Epochs = 10 (b) Epochs = 30 (c) Epochs = 50

Figure 4: 不同训练轮数下的词向量 t-SNE 聚类效果对比

3.3 Similarity Results

3.3.1 Epochs: 10

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	retail	rush	spouse	surprised	marriage
he	group	broad	byrd	reasons	unable
bank	a	the	tenders	monetary	in
responsibility	norwegian	neither	senator	giuliani	onto
eagerness	contentious	guber	pleasure	lilly	extensive

Table 7: Top 5 similar words for 10 epochs

3.3.2 Epochs: 20

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	rush	spouse	northwest	dialing	inclined
he	that	the	to	<unk>	of
bank	corp.	million	a	of	$
responsibility	congressional	suggests	legal	moreover	norwegian
eagerness	slipped	kkr	nathan	promotions	prohibition

Table 8: Top 5 similar words for 20 epochs

3.3.3 Epochs: 30

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	rush	northwest	spouse	retail	holdings
he	the	that	for	and	to
bank	a	operations	of	corp.	million
responsibility	congressional	norwegian	suggests	senator	player
eagerness	slipped	contentious	guber	nathan	about

Table 9: Top 5 similar words for 30 epochs

3.3.4 Epochs: 40

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	rush	spouse	franco	representatives	dialing
he	that	the	a	his	to
bank	corp.	million	of	officer	a
responsibility	congressional	suggests	legal	moreover	arena
eagerness	opportunities	incorporated	promotions	kkr	candlestick

Table 10: Top 5 similar words for 40 epochs

3.3.5 Epochs: 50

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	listening	recall	floor	peddling	he
he	it	had	no	have	that
bank	banking	securities	branch	loans	rating
responsibility	day-to-day	connections	responsible	wolf	shipyard
eagerness	simpler	poorest	unless	jokes	introduce

Table 11: Top 5 similar words for 50 epochs

3.4 Word Analogy Results

3.4.1 Epochs: 10

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	bridges	country	charges	raise	helmut
paris - france + italy	tested	assigned	cambria	killed	presidency
tokyo - japan + germany	years	ringing	berlin	lone	appropriate
doctor - hospital + teacher	requiring	sustain	no.	intelligence	middle-class

Table 12: Top 5 word analogy predictions for 10 epochs

3.4.2 Epochs: 20

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	went	pittsburgh	discovery	see	rainbow
paris - france + italy	nov.	prevented	presidency	tested	rogers
tokyo - japan + germany	stock	years	ringing	N	reached
doctor - hospital + teacher	conducted	serious	intelligence	instrumentation	liquidated

Table 13: Top 5 word analogy predictions for 20 epochs

3.4.3 Epochs: 30

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	went	discovery	charges	pittsburgh	bridges
paris - france + italy	tested	nov.	presidency	assigned	killed
tokyo - japan + germany	years	reached	ringing	N	berlin
doctor - hospital + teacher	intelligence	requiring	conducted	serious	liquidated

Table 14: Top 5 word analogy predictions for 30 epochs

3.4.4 Epochs: 40

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	pittsburgh	went	discovery	u.s.	rainbow
paris - france + italy	nov.	prevented	college	presidency	rogers
tokyo - japan + germany	ringing	stock	years	high	midsized
doctor - hospital + teacher	conducted	instrumentation	serious	press	manuel

Table 15: Top 5 word analogy predictions for 40 epochs

3.4.5 Epochs: 50

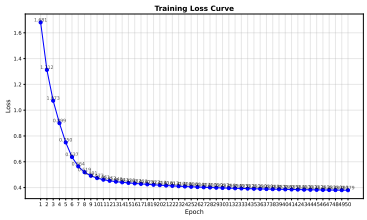
Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	pittsburgh	u.s.	rainbow	scientific	include
paris - france + italy	nov.	college	presidency	governments	killed
tokyo - japan + germany	ringing	stock	germans	shearson	midsized
doctor - hospital + teacher	conducted	iran-contra	manuel	mrs.	press

Table 16: Top 5 word analogy predictions for 50 epochs

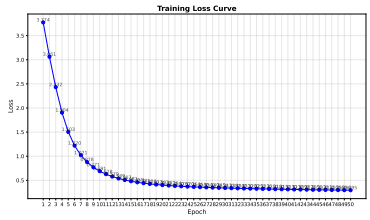
4 embed_size

- `learning_rate`: 0.001
- `embed_size`: 20, 100, 200, 500, 1000
- `epoch`: 50

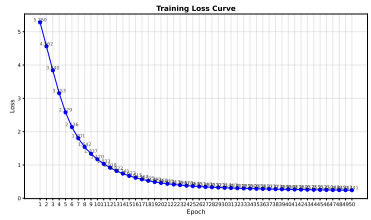
4.1 Training Curves



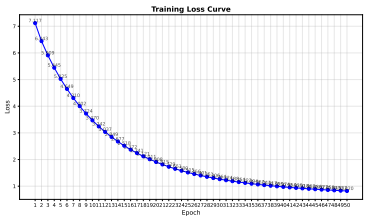
(a) Dim = 20



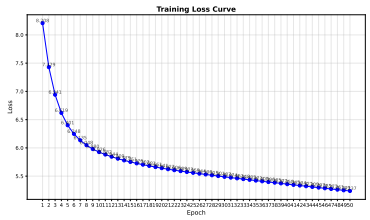
(b) Dim = 100



(c) Dim = 200



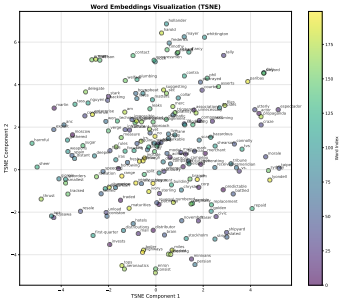
(d) Dim = 500



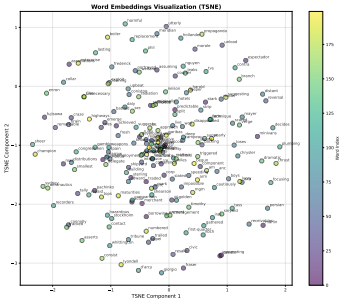
(e) Dim = 1000

Figure 5: 不同嵌入维度（Embedding Dimensions）下的训练曲线对比

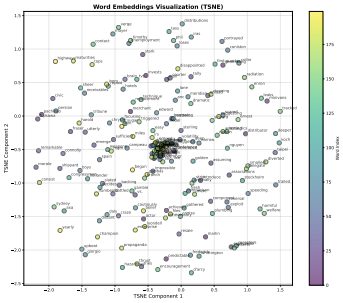
4.2 Visualizations



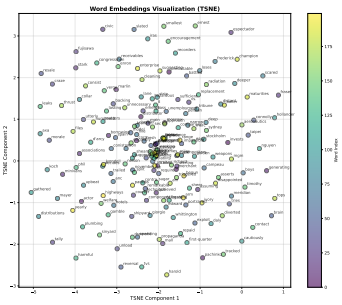
(a) Dim = 20



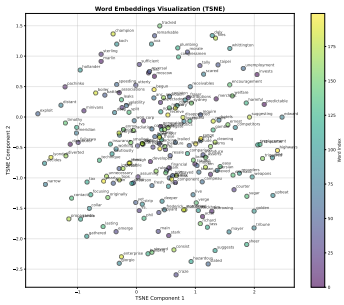
(b) Dim = 100



(c) Dim = 200



(d) Dim = 500



(e) Dim = 1000

Figure 6: 不同嵌入维度下的词向量 t-SNE 可视化对比

4.3 Similarity Results

4.3.1 Embedding Size: 20

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	dogs	bare-faced	unraveled	topiary	collectors
he	his	just	but	'm	ever
bank	conglomerate	traub	switzerland	daiwa	wachovia
responsibility	cia	contraceptive	webster	challenged	maximize
eagerness	fleet	confirm	attempt	airways	boom

Table 17: Top 5 similar words for embedding size 20

4.3.2 Embedding Size: 100

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	tighten	labeled	left	milton	publisher
he	that	mr.	his	who	is
bank	holding	officer	international	elected	investment
responsibility	funding	supreme	makes	nonexecutive	pay
eagerness	gauge	pleasure	carrying	opposes	saved

Table 18: Top 5 similar words for embedding size 100

4.3.3 Embedding Size: 200

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	listening	recall	floor	peddling	he
he	it	had	no	have	that
bank	banking	securities	branch	loans	rating
responsibility	day-to-day	connections	responsible	wolf	shipyard
eagerness	simpler	poorest	unless	jokes	introduce

Table 19: Top 5 similar words for embedding size 200

4.3.4 Embedding Size: 500

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	spielvogel	plunging	whitbread	projection	crushed
he	deposit	contributions	a.c.	magnified	objections
bank	chile	weaken	matching	have	script
responsibility	clout	conglomerate	millions	complex	during
eagerness	theatrical	suites	lbo	pinkerton	closely

Table 20: Top 5 similar words for embedding size 500

4.3.5 Embedding Size: 1000

Query Word	Top 1	Top 2	Top 3	Top 4	Top 5
dog	acknowledged	johns	design	gathered	circumstances
he	surge	reporters	unity	manufacturers	toward
bank	flat	january	effort	kabul	several
responsibility	pipeline	cars	convince	prestige	lost
eagerness	failing	seek	painted	becoming	speculated

Table 21: Top 5 similar words for embedding size 1000

4.4 Word Analogy Results

4.4.1 Embedding Size: 20

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	magna	h.	seagate	tucker	well
paris - france + italy	brussels	remics	frankfurt	spanish	kuala
tokyo - japan + germany	trading	weight	190-point	explosions	mercantile
doctor - hospital + teacher	wipe	custody	pot	attack	ringers

Table 22: Top 5 word analogy predictions for embedding size 20

4.4.2 Embedding Size: 100

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	practice	blockbuster	love	us	oas
paris - france + italy	raider	seoul	dayton	milan	friend
tokyo - japan + germany	prompting	intervened	reform	announcement	york
doctor - hospital + teacher	restrictions	she	socialist	figuring	spanish

Table 23: Top 5 word analogy predictions for embedding size 100

4.4.3 Embedding Size: 200

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	pittsburgh	u.s.	rainbow	scientific	include
paris - france + italy	nov.	college	presidency	governments	killed
tokyo - japan + germany	ringing	stock	germans	shearson	midsized
doctor - hospital + teacher	conducted	iran-contra	manuel	mrs.	press

Table 24: Top 5 word analogy predictions for embedding size 200

4.4.4 Embedding Size: 500

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	eliminate	casting	strategist	terminal	refunds
paris - france + italy	express	modernize	fair	road	conservatorship
tokyo - japan + germany	growing	container	's	accomplish	panamanian
doctor - hospital + teacher	venture-capital	perform	chooses	dire	aluminum

Table 25: Top 5 word analogy predictions for embedding size 500

4.4.5 Embedding Size: 1000

Analogy Task	Top 1	Top 2	Top 3	Top 4	Top 5
king - man + woman	diaper	brings	pa	bearings	flowers
paris - france + italy	evaluating	easy	medical	diego	machine
tokyo - japan + germany	statutory	prefers	bureaucrat	rowe	chartered
doctor - hospital + teacher	purchases	pachinko	p.m.	improved	removed

Table 26: Top 5 word analogy predictions for embedding size 1000

5 Analysis

Analysis

1. Word2Vec 的训练相对比较稳定，训练曲线在不同学习率的设置下都在稳定下降。
2. 从降维可视化的视角来看，**Word2Vec** 的过程是将原始在高维空间中随机分布的高维向量逐渐拉近或者拉远，形成团簇或者点簇（表征相关性）。
3. 选择不同维度的 `embed_size` 对高维数据的表征的表现不同，高维的向量训练损失更高、所需训练轮次长、并且从可视化的结果上最终不同 token 形成团簇的结构也不同。**适中的 `embed_size` (200) 可以一方面避免训练时间过长，另一方面也可以保证模型较好的捕捉相关性的核心信息**。